

Accruing evidence on benefits of adherence to the Mediterranean diet on health: an updated systematic review and meta-analysis.

BACKGROUND: The Mediterranean diet has long been reported to be protective against the occurrence of several different health outcomes. **OBJECTIVE:** We aimed to update our previous meta-analysis of published cohort prospective studies that investigated the effects of adherence to the Mediterranean diet on health status. **DESIGN:** We conducted a comprehensive literature search through electronic databases up to June 2010. **RESULTS:** The updated review process showed 7 prospective studies published in the past 2 y that were not included in the previous meta-analysis (1 study for overall mortality, 3 studies for cardiovascular incidence or mortality, 1 study for cancer incidence or mortality, and 2 studies for neurodegenerative diseases). These recent studies included 2 health outcomes not previously investigated (ie, mild cognitive impairment and stroke). The meta-analysis for all studies with a random-effects model that was conducted after the inclusion of these recent studies showed that a 2-point increase in adherence to the Mediterranean diet was associated with a significant reduction of overall mortality [relative risk (RR) = 0.92; 95% CI: 0.90, 0.94], cardiovascular incidence or mortality (RR = 0.90; 95% CI: 0.87, 0.93), cancer incidence or mortality (RR = 0.94; 95% CI: 0.92, 0.96), and neurodegenerative diseases (RR = 0.87; 95% CI: 0.81, 0.94). The meta-regression analysis showed that sample size was the most significant contributor to the model because it significantly influenced the estimate of the association for overall mortality. **CONCLUSION:** This updated meta-analysis confirms, in a larger number of subjects and studies, the significant and consistent protection provided by adherence to the Mediterranean diet in relation to the occurrence of major chronic degenerative diseases.